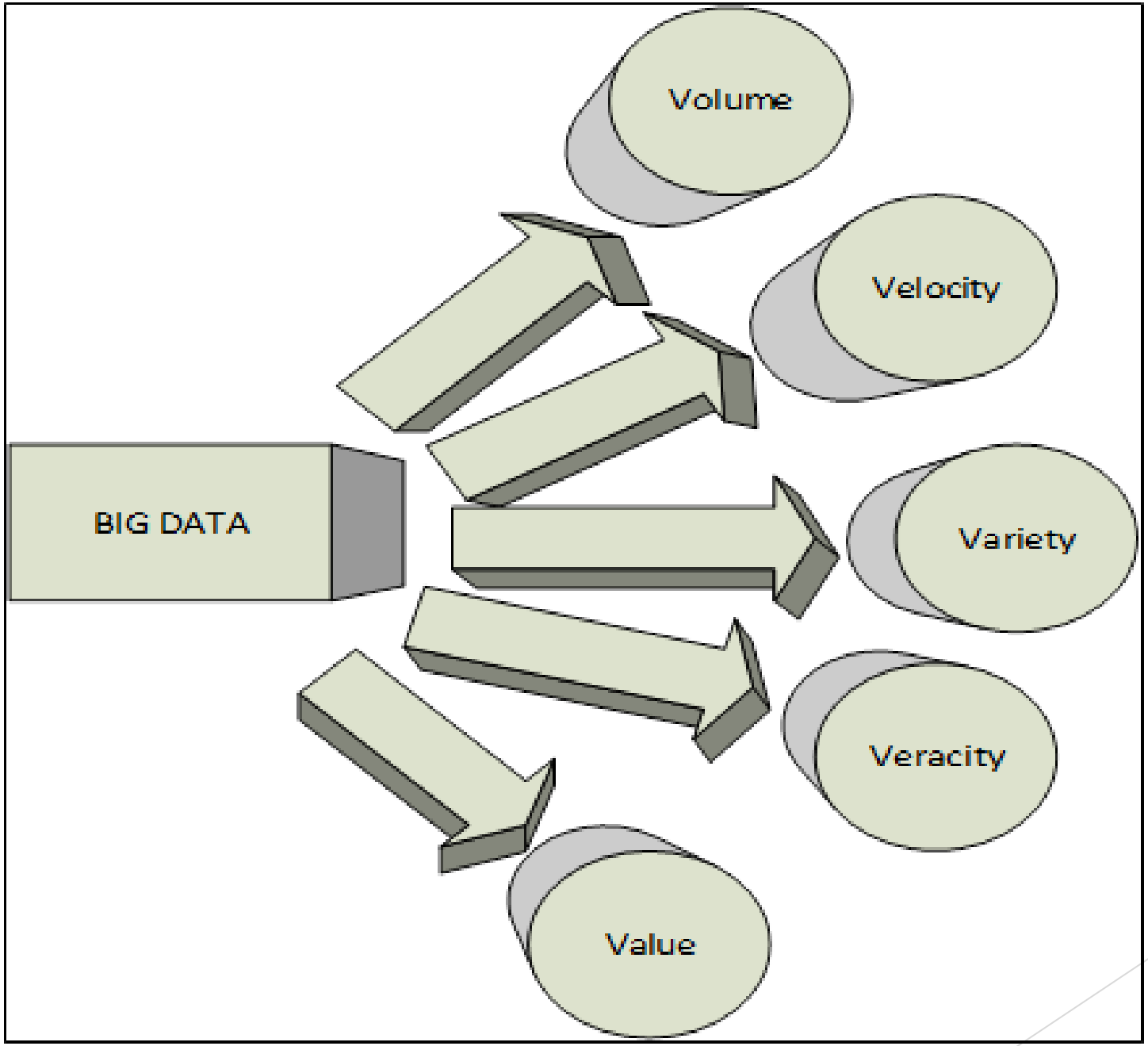
Big Data and Analytics

Twenty Sixth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Big Data Definition

- Big Data is a group of data from conventional and digital bases inside/outside the enterprise that characterizes a basis for continuing detection and analysis. (Lisa A 2013 *Big data marketing* (New Jersey: John Wiley & Sons, Inc.)
- Big Data is a quantity of data that is extremely relative and cannot be managed through the usual methods. (Bernard M 2016 *Big data in practice: How 45 successful companies used big data analytics to deliver extraordinary results* (New Jersey: John Wiley & Sons, Inc.)
- Big Data refers to extremely large and complex datasets—characterized by high volume, velocity, and variety—that exceed the capabilities of traditional data processing tools, requiring advanced methods for storage, analysis, and visualization to generate meaningful insights. (AI generated)


Volume
Velocity
BIG DATA
Variety
Veracity
Value

Interesting facts

- **WhatsApp**: Over **100 billion messages** are sent every single day. That's more than **1 million messages per second**!
- **YouTube**: Every minute, users upload **500+ hours of video**. That's more than **30,000 hours** every hour!
- **Google**: Processes over **8.5 billion searches** per day—that's about **100,000 searches every second**.
- **Instagram**: Users share **over 1,000 photos per second**.
- **Amazon**: Handles **hundreds of millions of transactions daily**, using Big Data to recommend what you'll probably buy next.

Growth of data

•By 2025, the world is expected to generate **over 180 zettabytes** of data.
(1 zettabyte = 10^{21} bytes)
A byte is a unit of data that is eight binary digits (bits) long.
•That's enough data to fill **23 trillion (10^{12})DVDs**—or stack DVDs from the Earth to the Moon **about 100 times**!
•About **90% of the world's data** was created **in just the last two years**.

Number	Scientific Notation	Number
Million	10^6	1,000,000
Billion	10^9	1,000,000,000
Trillion	10^{12}	1,000,000,000,000
Quadrillion	10^{15}	1,000,000,000,000,000
Quintillion	10^{18}	1,000,000,000,000,000,000

Unit	Decimal Value	Binary Equivalent (Approximate)
Byte	8 bits (1 character)	8 bits
Kilobyte (KB)	1,000 bytes	1,024 bytes
Megabyte (MB)	1,000,000 bytes	1,048,576 bytes
Gigabyte (GB)	1,000,000,000 bytes	1,073,741,824 bytes
Terabyte (TB)	1,000,000,000,000 bytes	1,099,511,627,776 bytes
Petabyte (PB)	1,000,000,000,000,000 bytes	1,125,899,906,842,624 bytes
Exabyte (EB)	1,000,000,000,000,000,000 bytes	1,152,920,390,729,171,120 bytes
Zettabyte (ZB)	1,000,000,000,000,000,000,000 bytes	1,180,591,620,717,411,208,384 bytes
Yottabyte (YB)	1,000,000,000,000,000,000,000,000 bytes	1,208,925,817,307,029,156,722,187, 168 bytes
Brontobyte (BB)	1,000,000,000,000,000,000,000,000, 000 bytes	1,220,501,287,654,504,376,949,544, 893,006,848 bytes

Need of Big Data in Manufacturing

- The following are the goals of the manufacturer:
  - Higher machining efficiency
  - Lower cost
  - Higher quality of the product
- The main aim of using big data in modern day industries is to ensure a cost efficient running without compromising the safety.
- The key performance indicators such as cutting force, cutting tool temperature, surface quality can be directly analysed using past data as well as real time data based on the controllable and uncontrollable variables.
- Proper implementation of big data in manufacturing can drastically reduce the product development time and other cost associated with it.

Objectives of Big Data Strategy and sources of data collection in manufacturing

- Accurately applying the machine learning techniques for prediction or classification.
- Draw relationship among the independent variables to gain knowledge for further analysis.

- The various sources used for capturing of machining data includes the following:
  - **Raw data can be collected from various instruments directly attached to the machine tools (dynamometer, current sensor, camera sensor)**
  - **The amount of cutting tool issued and other consumables can be collected from the store of an industry**
  - **The handwritten documents maintained by the machine tool operator while machining**

Challenges of Big Data

- The data generated by manufacturing industries are of lower volume when compared to Google or Cisco.
- Hence, Mourtzis (2016) called the whole paradigm as **INDUSTRIAL BIG DATA** and not **BIG DATA.**

Mourtzis, D., Vlachou, E. and Milas, N., 2016. Industrial Big Data as a result of IoT adoption in manufacturing. *Procedia cirp*, *55*, pp.290-295.

CAUSE	CHALLENGES
1. High dimensionality	A. Noise Accumulation
	B. Spurious Correlation
	C. Incidental Homogeneity

Challenges of Big Data (Continued)

CAUSE	CHALLENGES
2. High dimensionality and massive datasets	Heavy computational cost
3. Massive datsets	A. Heterogeneity
	B. Experimental Variation
	C. Statistical Biases

- Apart from the aforesaid challenges, reliability of the outcome of the data analysis and security of the machining data base from malicious attack are causes of concern
- Recognising the outliers so as consider the most important data for decision making.

Brief review on data driven manufacturing

- For data driven decision making, proper utilisation of data mining tools for knowledge discovery in databases (KDD) is necessary.
- The general steps involved in KDD includes the following steps (Mitra et al., 2002):

1. Knowledge about manufacturing
2. Collection of raw data
3. Data cleaning, pre-processing and transformation
4. Data integration
5. Selection of appropriate functions of data mining
6. Choosing the relevant algorithm to apply data mining
7. Data Mining
8. Inference and Visualization
9. Deployment of the obtained inference
10. Knowledge storage, reuse and integration into the manufacturing system

Mitra, S., Pal, S.K. and Mitra, P., 2002. Data mining in soft computing framework: a survey. *IEEE transactions on neural networks*, *13*(1), pp.3-14.

- In recent times, the focus has been more on the reliability and accuracy of the results obtained.
- Jian and Gao (2012) worked in order to introduce the prediction strategy for feedback control, production scheduling and optimization of industrial processes.

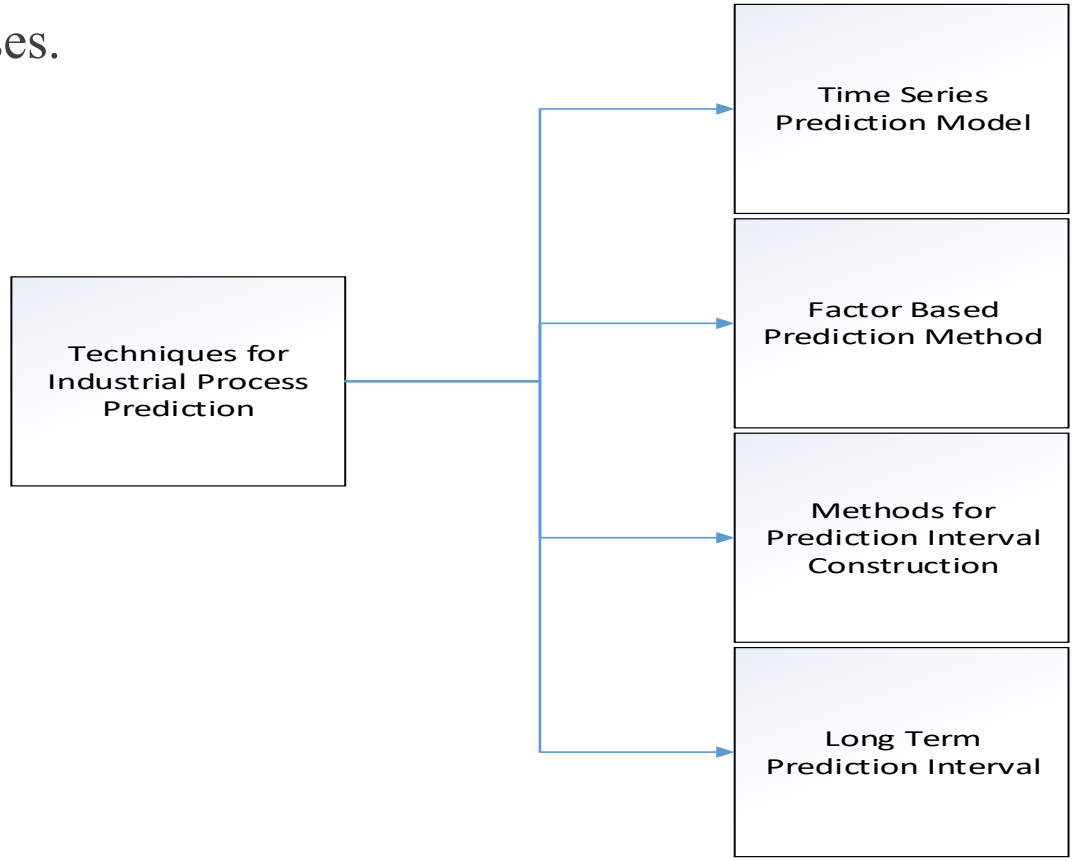


Figure Classification for techniques of industrial process prediction (Zhao et al., 2018)

- The following are the list of areas where data analytics have been applied in manufacturing by the researchers.

- Quality prediction of steel
- Condition monitoring based maintenance of smart system
- Power consumption in turning using a data driven neural network
- Data driven finite element method for cutting force estimation during turning operation
- Prediction of the rate of drilling in oil wells using random forest technique

Inference drawn from literature survey

- The focus of most of the researchers has been to show the capability in different sectors of engineering.
- The question of cost effectiveness arises for the implementation in modern day industries.
- There are very scarce papers in framework development which utilises the data from cloud in order to be implemented in real life situation.
- In addition to that there are numerous uncertain conditions arises in modern day industries which needs immediate attention of the researchers for data driven manufacturing.
- There are very few papers which provides answers to the following questions:
  - Is it justifiable to shift to data driven manufacturing when mechanistic models can still produce effective results?
  - What is the correct procedure of accommodating uncertainties such as tool wear, chatter and forced vibration in data driven prediction?

Hadoop

- Hadoop is an open-source, distributed computing framework designed for storing and processing large datasets. It's built on the principles of MapReduce, allowing parallel processing across a cluster of computers. Hadoop is used in various industries for tasks like data warehousing, log processing, and research.
- **Hadoop Distributed File System (HDFS):** HDFS is a distributed file system that stores data across multiple machines in a cluster. It's designed for high-throughput access and fault tolerance
- **MapReduce:** A programming model for processing data in parallel. It involves splitting data into small chunks, processing them in parallel (map), and then combining the results (reduce).
- **Hadoop YARN (Yet Another Resource Negotiator):** YARN is a resource management framework that provides a more flexible way to allocate resources across the cluster compared to the older MapReduce framework.

Hadoop Ecosystem

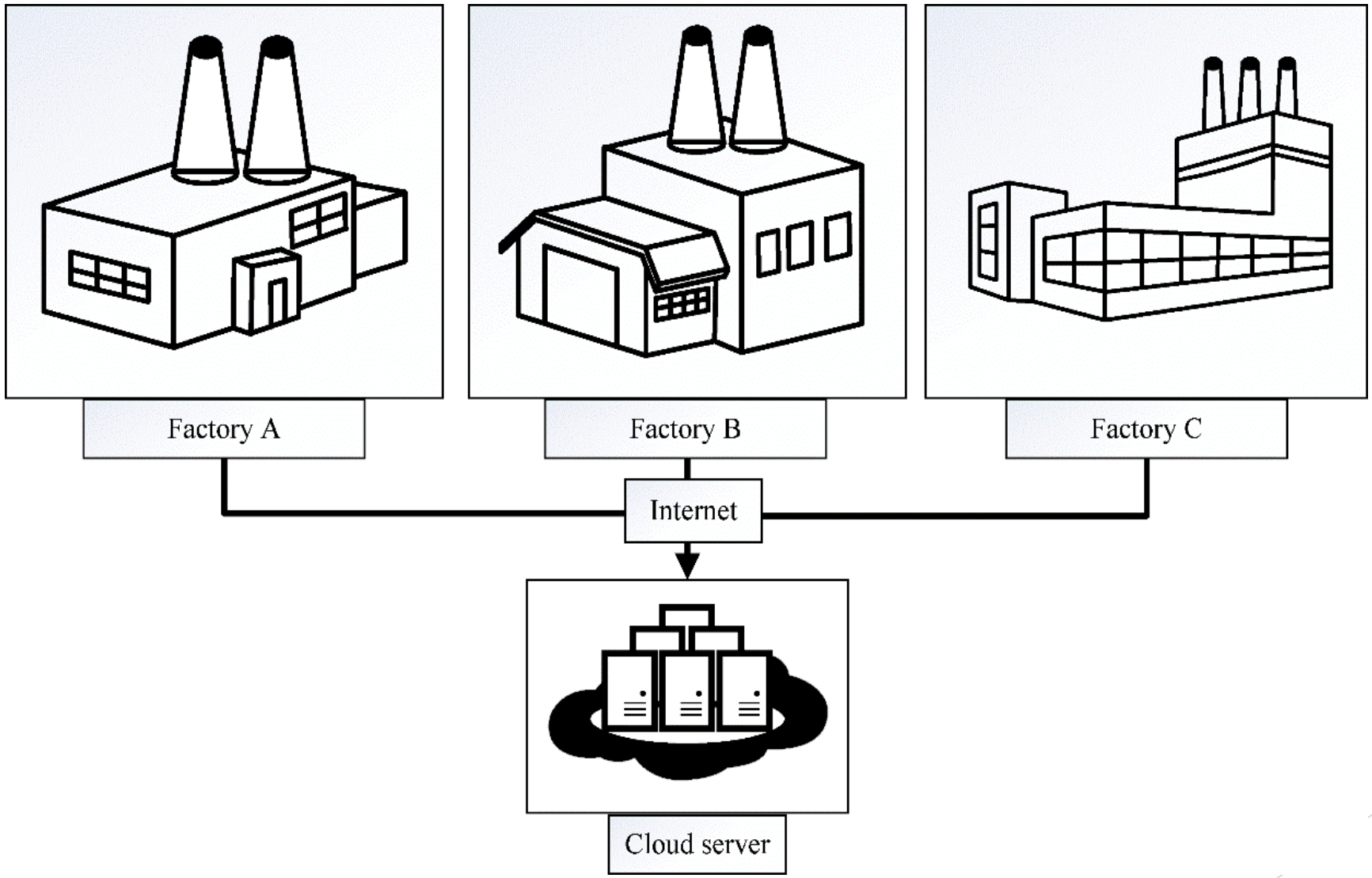
- **Hive:** A data warehouse system on top of Hadoop, allowing users to query data using SQL-like syntax.
- **Pig:** A high-level language for processing data in Hadoop, offering a more declarative approach than MapReduce.
- **Flume:** A tool for moving data into Hadoop from various sources.
- **Sqoop:** A tool for importing and exporting data between Hadoop and relational databases.
- **Oozie:** A workflow scheduler for managing and coordinating Hadoop jobs.
- **HBase:** A distributed, scalable NoSQL database on top of HDFS.

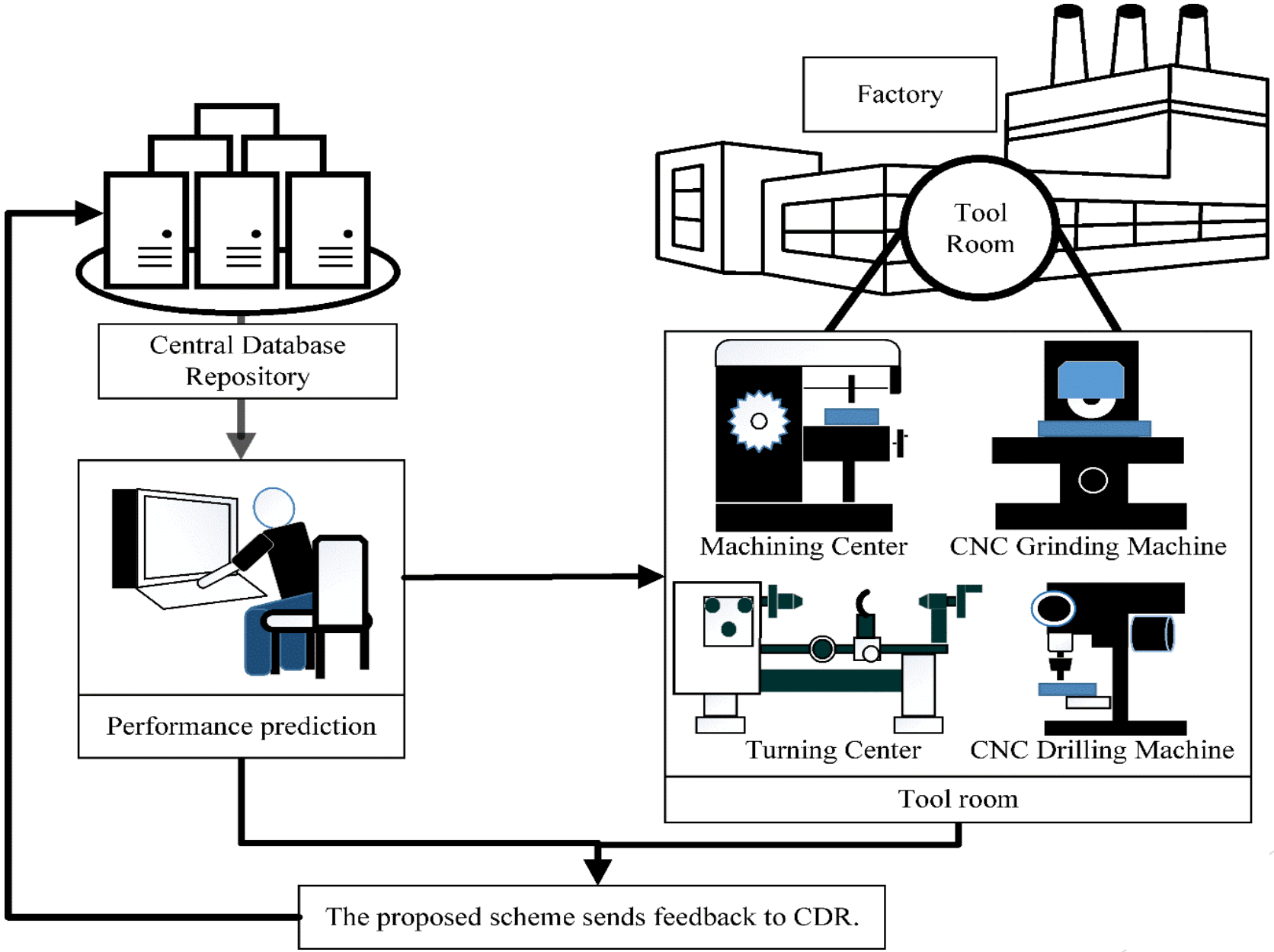
Benefits of Hadoop

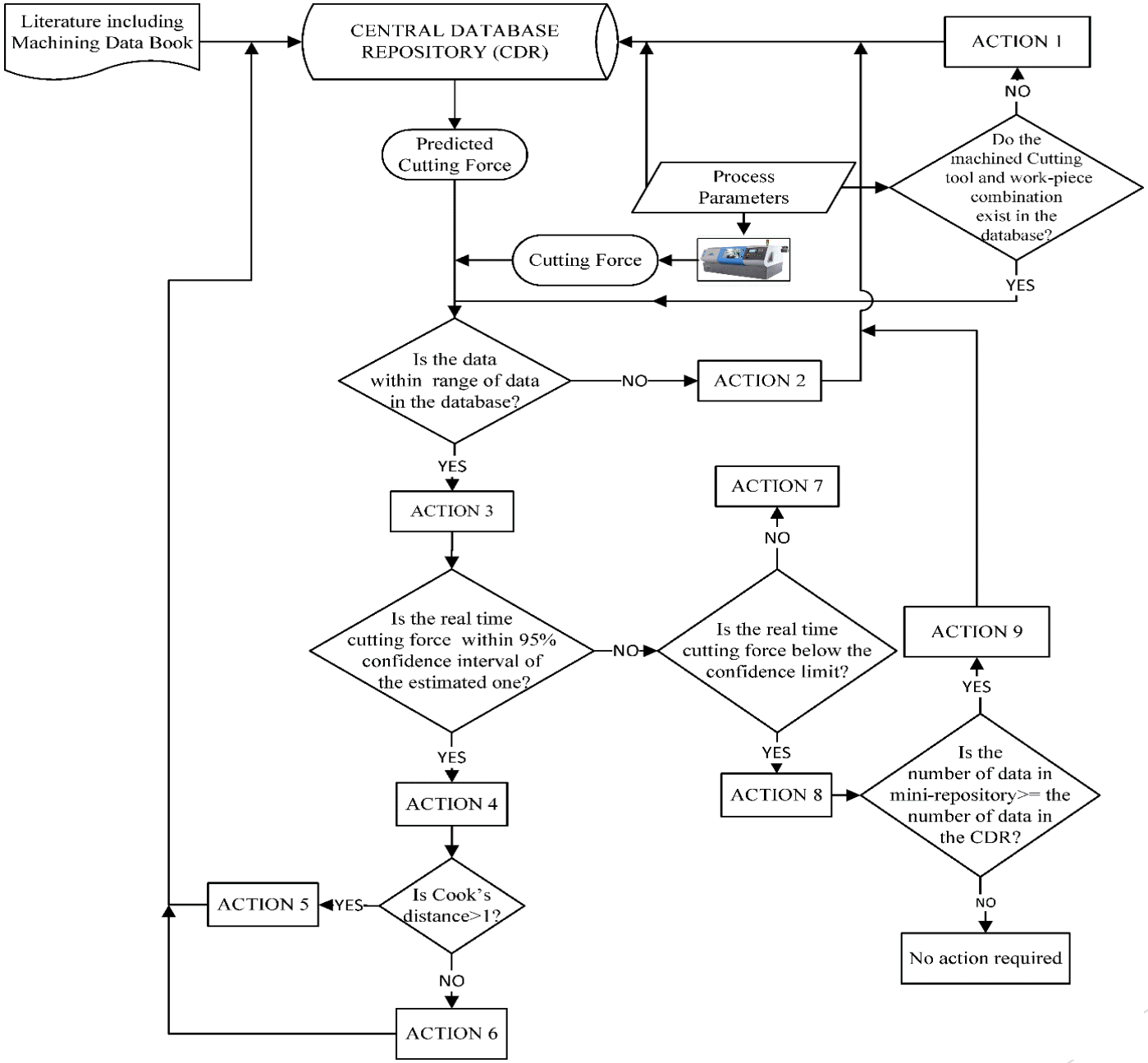
- **Scalability:** Hadoop can scale to handle massive amounts of data.
- **Cost-effectiveness:** It utilizes commodity hardware, making it more affordable than traditional data warehousing solutions.
- **Fault tolerance:** Hadoop is designed to automatically handle node failures, ensuring data integrity and availability.
- **Flexibility:** It can handle various data types and formats.
- **Data locality:** Computation is moved to the data, minimizing network traffic and improving processing speed.

Big data analytics

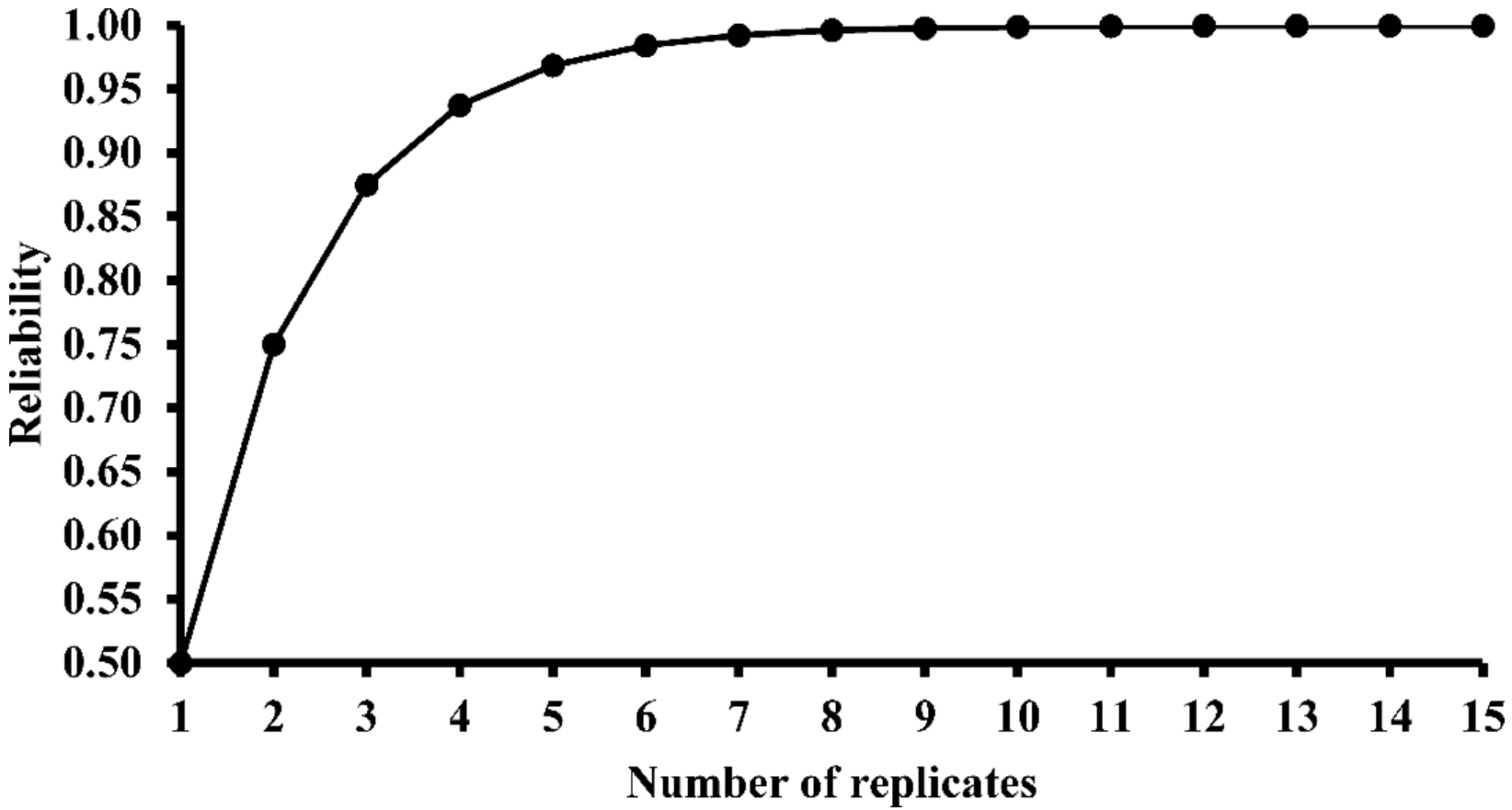
Analytics, at its core, is the process of examining data to discover significant patterns and extract meaningful insights. This involves using various techniques, including mathematical methods, statistics, and machine learning, to sift through large datasets and uncover new knowledge that might not be immediately apparent. In essence, analytics helps transform raw data into actionable information for decision-making.


Factory A
Factory B
Factory C
Internet
Cloud server


Factory
Tool Room
Central Database Repository
Machining Center
CNC Grinding Machine
Performance prediction
Turning Center
CNC Drilling Machine
Tool room
The proposed scheme sends feedback to CDR.


Literature including Machining Data Book
CENTRAL DATABASE REPOSITORY (CDR)
ACTION 1
NO
Predicted Cutting Force
Process Parameters
Do the machined Cutting tool and work-piece combination exist in the database?
Cutting Force
YES
Is the data within range of data in the database?
NO
ACTION 2
YES
ACTION 3
ACTION 7
NO
Is the real time cutting force within 95% confidence interval of the estimated one?
NO
Is the real time cutting force below the confidence limit?
ACTION 9
YES
YES
YES
ACTION 4
ACTION 8
Is the number of data in mini-repository>= the number of data in the CDR?
ACTION 5
YES
Is Cook's distance>1?
NO
No action required
NO
ACTION 6

Action Number	Task
1	Estimate the cutting force with all the existing models in the database repository and choose the one which is having the least error. Reduce the model error further by updating the estimation with a suitable correction factor and include the information in the prediction module for the new tool-work combination.
2	Extrapolated prediction is carried out with a caveat and the incoming process parameter along with the output details are included in the corresponding module of database repository. Assign a reliability of 0.5 to these data.
3	Extract the output from prediction model from Central Database Repository (CDR).
4	Calculate the Euclidean and Cook's distance.
5	Include the incoming process parameter and output details in the corresponding module of database repository. Assign a reliability of 0.5 to these data.
6	Improve the reliability of the nearest data (based on Euclidean distance) in the database repository.
7	Store the data in separate module for the analysis of tool wear or chatter.
8	Save the data in a mini repository.
9	Merge the data with the data in the CDR pertaining to the same tool-work combination, leaving aside outliers (error above $\mu+2\sigma$ limit). Fit new regression models.


Reliability
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Number of replicates

Cook’s distance

- A significant way to measure the influence of an observation is by using the Cook’s distance.
- Cook’s distance considers the residual of the output and the leverage of the point while estimating the level of influence.
- In other words, Cook’s distance combines the value of outlier and leverage for estimating the level of influence.
- It can be interpreted as a means to indicate the possible locations where it would be good to have a greater number of data points that can affect the future outcomes.
- Leverage: Measures how far an observation's explanatory variables (X values) are from the mean of all predictor variables. It indicates how much influence the observation has on the regression line based on its X values alone

Cook's distance and leverage

$$D_i = \frac{(y_i - y_{i,p})^2}{(m+1)s^2}\left\{\frac{h_i}{(1-h_i)^2}\right\},$$

$$h_i = \frac{1}{n} + \frac{(\ln V_p - \ln \overline{V})^2}{\sum_{i=1}^{n}\left(\ln V_i - \ln \overline{V}\right)^2} + \frac{(\ln f_p - \ln \overline{f})^2}{\sum_{i=1}^{n}\left(\ln f_i - \ln \overline{f}\right)^2} + \frac{(\ln d_p - \ln \overline{d})^2}{\sum_{i=1}^{n}\left(\ln d_i - \ln \overline{d}\right)^2}.$$

A data-entry is considered to be influential only when the value of Cook's distance exceeds one.

Cutting force with 95% prediction interval

$$= \exp\left\{ \ln \hat{F} \pm t(n-4, 5\%)\, s \sqrt{1 + \frac{1}{n} + \frac{(\ln V_p - \ln \bar{V})^2}{\sum_{i=1}^{n} \left(\ln V_i - \ln \bar{V} \right)^2} + \frac{(\ln f_p - \ln \bar{f})^2}{\sum_{i=1}^{n} \left(\ln f_i - \ln \bar{f} \right)^2} + \frac{(\ln d_p - \ln \bar{d})^2}{\sum_{i=1}^{n} \left(\ln d_i - \ln \bar{d} \right)^2}} \right\}$$

EFFECTIVE UTILIZATION OF DATA FOR ENHANCING THE PERFORMANCE OF MANUFACTURING

A Thesis Submitted in Partial Fulfillment of the Requirements
for the Degree of

DOCTOR OF PHILOSOPHY

by

Kaustabh Chatterjee
(Roll no 186103018)



Department of Mechanical Engineering
Indian Institute of Technology Guwahati
Guwahati 781039
INDIA

August 2023

Big Data in Machining

A report submitted for the
State of Art Seminar (SOAS)
by

Kaustabh Chatterjee
(Roll No. 186103018)
Supervisors
Prof. Uday Shanker Dixit
Dr. Jian Zhang

Department of Mechanical Engineering
Indian Institute of Technology Guwahati
Guwahati 781039

August 2019